

Лабораторная работа №14

Презентация

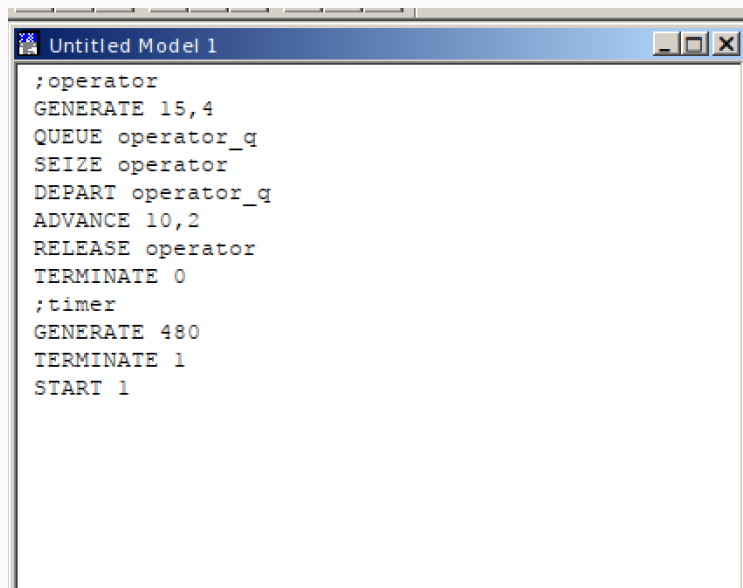
Дзахмишев Камбулат Заурович

10 мая 2025

Российский университет дружбы народов, Москва, Россия

Информация

- Дзахмишев Камбулат Заурович
- студент 3 курса
- Российский университет дружбы народов
- 1132221887@pfur.ru
- https://github.com/kzdzakhmishev/study_2024-2025_simmod

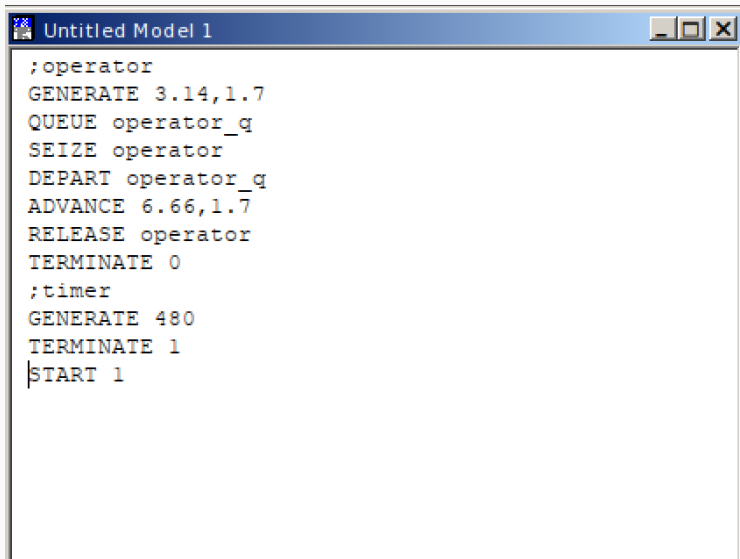


```
;operator
GENERATE 15,4
QUEUE operator_q
SEIZE operator
DEPART operator_q
ADVANCE 10,2
RELEASE operator
TERMINATE 0
;timer
GENERATE 480
TERMINATE 1
START 1
```

Выполнение лабораторной работы

Untitled Model 1.1.1 - REPORT													
GPSS World Simulation Report - Untitled Model 1.1.1													
cy66ota, max 10, 2025 12:40:29													
START TIME		END TIME		BLOCKS	FACILITIES	STORAGES							
0.000		480.000		9	1	0							
NAME		VALUE											
OPERATOR		10001.000											
OPERATOR_Q		10000.000											
LABEL	LOC	BLOCK TYPE	ENTRY	COUNT	CURRENT	COUNT	RETRY						
	1	GENERATE	22		0		0						
	2	QUEUE	22		0		0						
	3	SEIZE	22		0		0						
	4	DEPART	22		0		0						
	5	ADVANCE	22		1		0						
	6	RELEASE	21		0		0						
	7	TERMINATE	21		0		0						
	8	GENERATE	1		0		0						
	9	TERMINATE	1		0		0						
FACILITY	ENTRIES	UTIL.	AVE. TIME	AVAIL.	OWNER	PEND	INTER	RETRY	DELAY				
OPERATOR	22	0.629	9.589	1	22	0	0	0	0				
QUEUE	MAX CONT.	ENTRY	ENTRY(0)	AVE. CONT.	AVE. TIME	AVE. (-0)	RETRY						
OPERATOR_Q	1	0	22	21	0.001	0.021	0.671	0					
FEC XN	PRI	BDT	ASSEM	CURRENT	NEXT	PARAMETER	VALUE						
23	0	489.786	23	5	6								
24	0	496.081	24	0	1								
25	0	960.000	25	0	8								

Рис. 2: Отчёт по заявкам.

A screenshot of a text editor window titled "Untitled Model 1". The window has a standard Windows-style title bar with minimize, maximize, and close buttons. The text inside the editor is as follows:

```
;operator  
GENERATE 3.14,1.7  
QUEUE operator_q  
SEIZE operator  
DEPART operator_q  
ADVANCE 6.66,1.7  
RELEASE operator  
TERMINATE 0  
;timer  
GENERATE 480  
TERMINATE 1  
|START 1
```

Выполнение лабораторной работы

суббота, мая 10, 2025 12:49:19

START TIME	END TIME	BLOCKS	FACILITIES	STORAGES
0.000	480.000	9	1	0

NAME	VALUE
OPERATOR	10001.000
OPERATOR_Q	10000.000

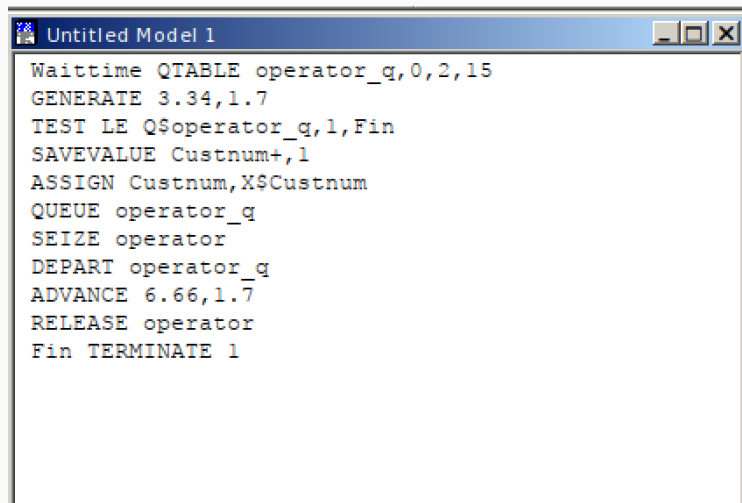
LABEL	LOC	BLOCK TYPE	ENTRY COUNT	CURRENT COUNT	RETRY
	1	GENERATE	152	0	0
	2	QUEUE	152	82	0
	3	SEIZE	70	0	0
	4	DEPART	70	0	0
	5	ADVANCE	70	1	0
	6	RELEASE	69	0	0
	7	TERMINATE	69	0	0
	8	GENERATE	1	0	0
	9	TERMINATE	1	0	0

FACILITY	ENTRIES	UTIL.	AVE. TIME	AVAIL.	OWNER	PEND	INTER	RETRY	DELAY
OPERATOR	70	0.991	6.796	1	71	0	0	0	82

QUEUE	MAX	CONT.	ENTRY	ENTRY(0)	AVE.CONT.	AVE.TIME	AVE.(-0)	RETRY
OPERATOR_Q	82	82	152	1	39.096	123.461	124.279	0

FEC	XN	PRI	BDT	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
	71	0	480.405	71	5	6		
	154	0	483.330	154	0	1		
	155	0	960.000	155	0	8		

Рис. 4: Отчёт по модели из задания.



```
Waittime QTABLE operator_q,0,2,15
GENERATE 3.34,1.7
TEST LE Q$operator_q,1,Fin
SAVEVALUE Custnum+,1
ASSIGN Custnum,X$Custnum
QUEUE operator_q
SEIZE operator
DEPART operator_q
ADVANCE 6.66,1.7
RELEASE operator
Fin TERMINATE 1
```

Рис. 5: Модель гистограммы распределения заявок.

Выполнение лабораторной работы

START TIME															
0.000															
END TIME		BLOCKS		FACILITIES		STORAGES									
353.895		10		1		0									
NAME					VALUE										
CUSTNUM					10002.000										
FIN					10.000										
OPERATOR					10003.000										
OPERATOR_Q					10001.000										
WAITTIME					10000.000										
LABEL															
LOC		BLOCK TYPE		ENTRY COUNT		CURRENT COUNT		RETRY							
1		GENERATE		102		0		0							
2		TEST		102		0		0							
3		SAVEVALUE		55		0		0							
4		ASSIGN		55		0		0							
5		QUEUE		55		1		0							
6		SEIZE		54		1		0							
7		DEPART		53		0		0							
8		ADVANCE		53		0		0							
9		RELEASE		53		0		0							
FIN		10 TERMINATE		100		0		0							
FACILITY															
ENTRIES		UTIL.		AVE. TIME		AVAIL. OWNER		PEND INTER							
54		0.987		6.470		1		98							
0		0		0		0		0							
1		1		1		1		1							
QUEUE															
MAX CONT.		ENTRY ENTRY(0)		AVE.CONT.		AVE.TIME		AVE.(-0)							
2		2 55		1		1.652		10.628							
0		0		0		0		0							
1		1		1		1		1							
TABLE															
MEAN		STD.DEV.		RANGE		RETRY FREQUENCY		CUM. %							
10.709		2.702		0		0		0							
0.000		0.000		0.000		1		1.89							
0.000		0.000		2.000		0		1.89							
2.000		2.000		4.000		1		3.77							

Рис. 6: Отчёт по модели.

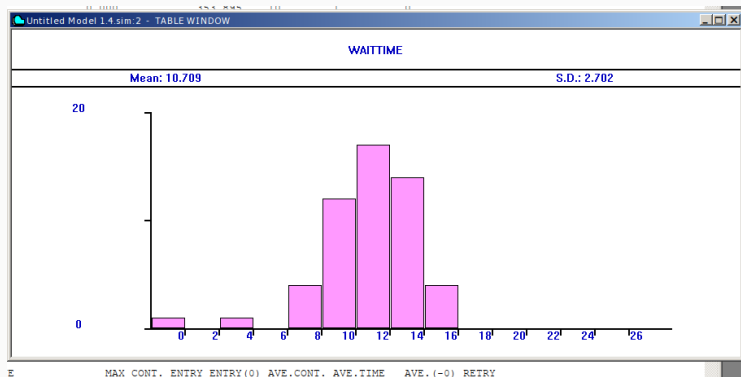
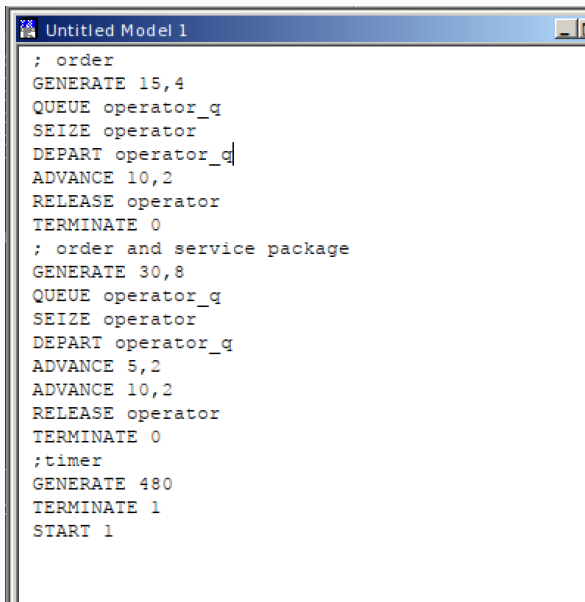


Рис. 7: Гистограмма.



```
; order
GENERATE 15,4
QUEUE operator_q
SEIZE operator
DEPART operator_q
ADVANCE 10,2
RELEASE operator
TERMINATE 0

; order and service package
GENERATE 30,8
QUEUE operator_q
SEIZE operator
DEPART operator_q
ADVANCE 5,2
ADVANCE 10,2
RELEASE operator
TERMINATE 0

;timer
GENERATE 480
TERMINATE 1
START 1
```

Выполнение лабораторной работы

суббота, мая 10, 2025 13:02:17

START TIME	END TIME	BLOCKS	FACILITIES	STORAGES
0.000	480.000	17	1	0

NAME	VALUE
OPERATOR	10001.000
OPERATOR_Q	10000.000

LABEL	LOC	BLOCK TYPE	ENTRY COUNT	CURRENT	COUNT	RETRY
	1	GENERATE	32		0	0
	2	QUEUE	32		4	0
	3	SEIZE	28		0	0
	4	DEPART	28		0	0
	5	ADVANCE	28		1	0
	6	RELEASE	27		0	0
	7	TERMINATE	27		0	0
	8	GENERATE	15		0	0
	9	QUEUE	15		3	0
	10	SEIZE	12		0	0
	11	DEPART	12		0	0
	12	ADVANCE	12		0	0
	13	ADVANCE	12		0	0
	14	RELEASE	12		0	0
	15	TERMINATE	12		0	0
	16	GENERATE	1		0	0
	17	TERMINATE	1		0	0

FACILITY	ENTRIES	UTIL.	AVE. TIME	AVAIL.	OWNER	PEND	INTER	RETRY	DELAY
OPERATOR	40	0.947	11.365	1	42	0	0	0	7

QUEUE	MAX CONT.	ENTRY	ENTRY(0)	AVE. CONT.	AVE. TIME	AVE. (-0)	RETRY
OPERATOR_Q	8	7	47	2	3.355	34.261	35.784 0

FEC	XN	PRI	BDT	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
42	0		487.825	42	5	6		
50	0		493.164	50	0	1		
49	0		499.562	49	0	8		
51	0		960.000	51	0	16		

```
; order
GENERATE 15,4
QUEUE operator_q
|SEIZE operator
DEPART operator_q
ADVANCE 10,2
TRANSFER 0.3,noextr,extr
extr ADVANCE 5,2
noextr RELEASE operator
TERMINATE 0
;timer
GENERATE 480
TERMINATE 1
START 1
```

Выполнение лабораторной работы

Untitled Model 1.6.1 - REPORT

GPSS World Simulation Report - Untitled Model 1.6.1

cy66eta, max 10, 2025 13:10:12

START TIME	END TIME	BLOCKS	FACILITIES	STORAGES
0.000	480.000	11	1	0

NAME	VALUE
EXTR	7.000
NOEXTR	8.000
OPERATOR	10001.000
OPERATOR_Q	10000.000

LABEL	LOC	BLOCK TYPE	ENTRY COUNT	CURRENT COUNT	RETRY
	1	GENERATE	33	0	0
	2	QUEUE	33	0	0
	3	SEIZE	33	0	0
	4	DEPART	33	0	0
	5	ADVANCE	33	0	0
	6	TRANSFER	33	0	0
EXTR	7	ADVANCE	8	1	0
NOEXTR	8	RELEASE	32	0	0
	9	TERMINATE	32	0	0
	10	GENERATE	1	0	0
	11	TERMINATE	1	0	0

FACILITY	ENTRIES	UTIL.	AVE. TIME	AVAIL.	OWNER	PEND	INTER	RETRY	DELAY
OPERATOR	33	0.766	11.146	1	34	0	0	0	0

QUEUE	MAX CONT.	ENTRY	ENTRY (0)	AVE. CONT.	AVE. TIME	AVE. (-0)	RETRY	
OPERATOR_Q	1	0	33	25	0.054	0.781	3.220	0

FEC	XN	PRI	BDT	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
34	0		482.925	34	7	8		
35	0		487.726	35	0	1		
36	0		960.000	36	0	10		



Untitled Model 1

```
operator STORAGE 4  
GENERATE 5,2  
QUEUE operator_q  
ENTER operator,1  
DEPART operator_q  
ADVANCE 10,2  
LEAVE operator,1  
TERMINATE 0  
;timer  
GENERATE 480  
TERMINATE 1  
START 1
```

Выполнение лабораторной работы

Untitled Model 1.7.1 - REPORT													
GPSS World Simulation Report - Untitled Model 1.7.1													
cy666ora, max 10, 2025 13:14:14													
START TIME		END TIME		BLOCKS	FACILITIES	STORAGES							
0.000		480.000		9	0	1							
NAME		VALUE											
OPERATOR		10000.000											
OPERATOR_Q		10001.000											
LABEL	LOC	BLOCK TYPE	ENTRY	COUNT	CURRENT	COUNT	RETRY						
1		GENERATE	93		0	0	0						
2		QUEUE	93		0	0	0						
3		ENTER	93		0	0	0						
4		DEPART	93		0	0	0						
5		ADVANCE	93		2	0	0						
6		LEAVE	91		0	0	0						
7		TERMINATE	91		0	0	0						
8		GENERATE	1		0	0	0						
9		TERMINATE	1		0	0	0						
QUEUE	MAX CONT.	ENTRY	ENTRY(0)	AVE. CONT.	AVE. TIME	AVE. (-0)	RETRY						
OPERATOR_Q	1	0	93	93	0.000	0.000	0.000	0					
STORAGE	CAP.	REM.	MIN.	MAX.	ENTRIES	AVL.	AVE. C.	UTIL.	RETRY DELAY				
OPERATOR	4	2	0	4	93	1	1.926	0.482	0 0				
FEC	XN	PRI	BDT	ASSEM	CURRENT	NEXT	PARAMETER	VALUE					
95	0		480.457	95	0	1							
93	0		482.805	93	5	6							
94	0		483.473	94	5	6							
96	0		960.000	96	0	8							


```
operator STORAGE 4
GENERATE 5,2
TEST LE Q$operator_q,2
QUEUE operator_q
ENTER operator,1
DEPART operator_q
ADVANCE 30,2
LEAVE operator,1
TERMINATE 0
;timer
GENERATE 480
TERMINATE 1
START 1
```

Выполнение лабораторной работы

Untitled Model 1.8.1 - REPORT																
GPSS World Simulation Report - Untitled Model 1.8.1																
суббота, мая 10, 2025 12:18:07																
START TIME		END TIME		BLOCKS	FACILITIES		STORAGES									
0.000		480.000		10	0		1									
NAME		VALUE														
OPERATOR		10000.000														
OPERATOR_Q		10001.000														
LABEL	LOC	BLOCK TYPE	ENTRY COUNT		CURRENT	COUNT	RETRY									
	1	GENERATE	94			27	0									
	2	TEST	67			0	0									
	3	QUEUE	67			3	0									
	4	ENTER	64			0	0									
	5	DEPART	64			0	0									
	6	ADVANCE	64			4	0									
	7	LEAVE	60			0	0									
	8	TERMINATE	60			0	0									
	9	GENERATE	1			0	0									
	10	TERMINATE	1			0	0									
QUEUE																
OPERATOR_Q	3	3	67	4	2.701	19.347	20.576	27								
STORAGE																
OPERATOR	4	0	0	4	64	1	3.885	0.971	0	3						
FEC	XN	PRI	BDT	ASSEM	CURRENT	NEXT	PARAMETER	VALUE								
96	0		480.736	96	0	1										
62	0		491.784	62	6	7										
63	0		491.929	63	6	7										
64	0		495.070	64	6	7										
65	0		499.648	65	6	7										
97	0		960.000	97	0	9										

В ходе данной лабораторной работы получилось смоделировать несколько разных моделей поступления заказов и заявок.